

LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE
Mid Term Examinations, 2024

Paper Title: Introduction to Management

Class: BSCS

Session: 2023-2027

Semester: 3rd

Total Time: 2 hours

Total Marks: 30

SUBJECTIVE SECTION

095

Expanding into a New Market

Background: A medium-sized electronics company, "Tech Innovations," has been performing well in its local market. The company now faces a strategic decision: whether to expand into an international market. The management team is considering three options: expanding into a neighboring country, venturing into a distant but high-potential market, or staying local and focusing on product diversification.

Questions:

1. What steps should the management follow in the decision-making process to ensure the best decision is made?
2. How would you apply the rational decision-making model to this case?
3. What factors should be considered when evaluating the risks and potential rewards of each option?
4. How could bounded rationality or satisficing affect the decision in this scenario?
5. If you were the CEO, what decision would you make, and why?

Q2. Consider a business that you frequent (for example, a restaurant or coffee shop) and review the six aspects of the external environment discussed in the chapter. Create a list of factors in the external environment that could affect the management of the business you selected.

Q3. What is bureaucracy? Do bureaucracies still exist today?

Lahore College for Women University

Department: **Computer Science**

Paper: **Applied Physics**

Total Time: **1hours, 20mintues**

Paper Type: **Subjective Type**

Total Marks: **20**

095

Attempt all the questions.

Q 2. (a) There are two types of materials explain their elastic and inelastic limits also express what type of stress and strain is produced in the bodies with their young modulus expression.

(b) If a body having mass m and velocity v then what will be the momentum and the force applied on that body?

(c) There are three different displacements which a car covered in 20mintues are as $d_1 = 6m$, $d_2 = 8m$ and $d_3 = 5m$ with the following angles $\theta_1 = 40^\circ$, $\theta_2 = 30^\circ$ and $\theta_3 = 0^\circ$ respectively. Now find the net displacement in terms of their components.

Q 3. (a) If a body moving under the influence of gravity, then explain the Newtons law which is applied on that body

(b) A rocket is moving with some speed and make an angle θ with the earth, explain the flight of the rocket along with the path which is followed by the rocket and its maximum and minimum range.

(c) Two individuals A and B sliding a motor of weight 225kg with a displacement d of magnitude 8.50m. the push F_1 of A is 12N at an angle of 30° downward, while the pull F_2 of B is 10N at 40° above the horizontal. What is the net work done on the motor by force F_1 and F_2 during the displacement d ?

Lahore College for Women University

Final Term Examination 2024

Department of Computer Science

Class: BS- II

Semester III

(Session 2023-2027)

Course Code: GEDU-205

Marks: 50

Paper Title: Applied Physics(subjective type)

Time Allowed: 2.40 minutes

Note: Attempt all the questions.

Q3. Define Electric potential energy. Explain the potential due to an electric dipole.

/10

Q4. (a) What is capacitance? Derive the expression of C_{eq} capacitor in series combination. /5

(b) what are the effects of electric current? /5

Q5. (a) What is Faraday Law of Induction? /5

(b) What is ammeter? Explain the working principle of ammeter. /5

Q6.(a) What are semiconductors? Explain their types and uses. /5

(b) What are dielectric constants? /5

$$Q = CV \\ V = IR$$

$$P = I^2 R \quad \Delta V = ?$$

Q95

$$U = q\Delta V$$

$$\Delta V = Q/C$$

$$Q = C\Delta V \\ Q/C = \Delta V$$

$$Q = CV$$

$$\Delta V =$$

$$Q/C$$

$$Q = CV$$

28-11-2024

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LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE

Final Term Examinations, 2024

Paper Title: Introduction to Management

Class: BSCS

Session: 2023-2027

Semester: III

Total marks: 50

Total Time: 3 Hours

Date: 3/1/25

095

SUBJECTIVE

Sr. No.	Questions	CLO	PLO	Taxonomy level	Marks
1	Define groups and the stages of group development.	1	1	C2	15
2	How do goal setting, reinforcement, and equity theories explain employee motivation?	2	2	C1	15
3	A NEW KIND OF STRUCTURE Admit it. Sometimes the projects you're working on (school, work, or both) can get pretty boring and monotonous. Wouldn't it be great to have a magic button you could push to get someone else to do that boring, time-consuming stuff? At Pfizer, that "magic button" is a reality for a large number of employees. As a global pharmaceutical company, Pfizer is continually looking for ways to help employees be more efficient and effective. The company's senior director of organizational effectiveness found that the "Harvard MBA staff we hired to develop strategies and innovate were instead Googling and making Power Points." Indeed, internal studies conducted to find out just how much time its valuable talent was spending	3	2	C3	20


12/12/2024

on menial tasks was startling. The average Pfizer employee was spending 20 percent to 40 percent of his or her time on support work (creating documents, typing notes, doing research, manipulating data, scheduling meetings) and only 60 percent to 80 percent on knowledge work (strategy, innovation, networking, collaborating, critical thinking). And the problem wasn't just at lower levels. Even the highest-level employees were affected. Take, for instance, David Cain, an executive director for global engineering. He enjoys his job—assessing environmental estate risks, managing facilities, and controlling a multi million-dollar budget. But he didn't so much enjoy having to go through spreadsheets and put together Power-points. Now, however, with Pfizer's "magic button," those tasks are passed off to individuals outside the organization.

Just what is this "magic button?" Originally called the Office of the Future (OOF), the renamed Pfizer Works allows employees to shift tedious and time-consuming tasks with the click of a single button on their computer desktop. They describe what they need on an online form, which is then sent to one of two Indian service outsourcing firms. When a request is received, a team member in India calls the Pfizer employee to clarify what's needed and by when. The team member then e-mails back a cost specification for the requested work. If the Pfizer employee decides to proceed, the costs involved are charged to the employee's department. About this unique arrangement, Cain said that he relishes working with what he prefers to call his "personal consulting organization."

The number (66,500) illustrates just how beneficial PfizerWorks has been for the company. That's the number of work hours estimated to have been saved by employees who've used PfizerWorks. What about Joe Cain's experiences? When he gave the Indian team a complex

project researching strategic actions that worked when consolidating company facilities, the team put the report together in a month, something that would have taken him six months to do alone. He says, "Pfizer pays me not to work tactically, but to work strategically."

Discussion Questions:

1. Describe and evaluate what Pfizer is doing with its PfizerWorks.
2. What structural implications—good and bad—does this approach have? (Think in terms of the six organizational design elements.)
3. Do you think this arrangement would work for other types of organizations? Why or why not? What types of organizations might it also work for?
4. What role do you think organizational structure plays in an organization's efficiency and effectiveness? Explain.

- 1) departmentalization
- 2) work specialization
- 3) decentralization/centralization
- 4) span of control
- 5) chain of command
- 6) formalization.

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12/12/2024

Lahore College for Women University
Department of Computer Science
Final Term Examination

Course Title: Data Structures and Algorithms
 Semester: III

Fall Term 2024
 Session: 2023 to 2027

Max Marks: 100

Total Time: 3hrs

Students Name: Roshan

Roll No: 095

Note: Cutting and rubbing is not allowed. Only use black and blue ink to solve this paper. Attach question papers in front of answer sheet.

Subjective (Total Marks: 40)

Note: Attempt all questions. All Questions carry equal marks.

Q#2 Given a 5 element queue Q (from front to back: 1, 3, 5, 7, 9), and an empty stack S, remove the elements one-by-one from Q and insert them into S, then remove them one-by-one from S and re-insert them into Q. The queue now looks like (from front to back). Explain each step. (10)

ind ↑

1 3 5 7 9

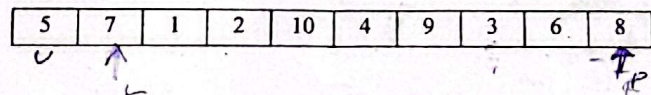
9
7
5
3
1

9
7
5
3
1

Q#2 What is closed hashing, how collisions will be resolved in it write an algorithm that insert N integer data values into a hash table and then searches a specific value from the hash table. Take an array of size N as data structure. (10)

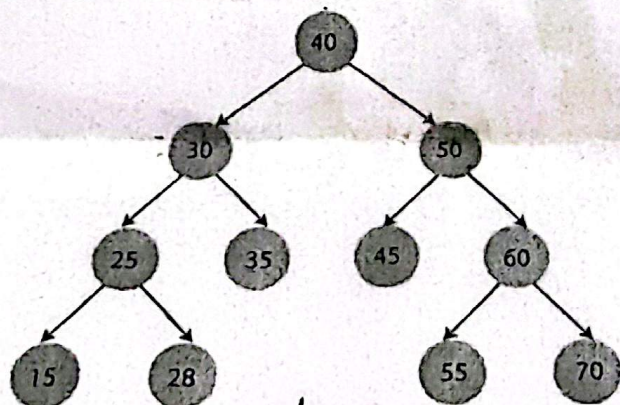
9 7 5 3 1

Q#3 Trace the process to sort the following array in ascending order using Quick sort algorithm. Also describe its algorithm and time complexity while ($U \neq P$ & $I \leq P$) swap(U & I) 5- (10)



Q # 4 What are different traversing operation perform on Binary Search Trees. Apply inorder traversing on the following binary tree and write its sequence and algorithm using stacks as data structure. (10)

index = elem's no. of elem.



5 3 1 2 10 4 9 7 6 8
 5 3 1

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 NLR
 LRN

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LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE

Final Term Examination 2025

Class: BSCS

Semester: III

Course Title: Calculus and Analytical Geometry

Course Code: GEDU-202

Max Marks: 50

Time allowed: 3hrs

Note: Attempt all questions

	Attempt all long Questions:	
Q#1(a)	Find the formulas for $f \circ g$ and $g \circ f$, and state the domain of the composition, if $f(x) = \frac{1+x}{1-x}$, $g(x) = \frac{x}{1-x}$.	4
(b)	Find the formula for the inverse of $f(x) = \sqrt{3x-2}$ with x as an independent variable and state the domain of $f^{-1}(x)$.	4
(c)	Let $f(x) = \begin{cases} x-1, & x \leq 3 \\ 3x-7, & x > 3 \end{cases}$. Also find $\lim_{x \rightarrow 3} f(x)$.	3
Q#2(a)	Evaluate the following limits. (i) $\lim_{x \rightarrow +\infty} \frac{1-e^x}{1+e^x}$, (ii) $\lim_{h \rightarrow 0} \frac{\sin h}{1-\cos h}$, (iii) $\lim_{x \rightarrow -\infty} \frac{4x-1}{2x+5}$, (iv) $\lim_{x \rightarrow +\infty} \left(1 + \frac{3}{x}\right)^{-x}$.	10
Q#3(a)	Define continuity, what can you say about the continuity of the function $f(x) = \sqrt{9-x^2}$?	5
(b)	Find the constants k and m , if possible, that will make the function f continuous everywhere. $f(x) = \begin{cases} x^2 + 5, & x > 2 \\ m(x+1) + k, & -1 < x \leq 2 \\ 2x^3 + 7, & x \leq -1 \end{cases}$	5
Q#4(a)	Show that $y = x \sin x$ is a solution to the differential equation	3

Q # 1 Answer the following?

(marks 5*5=25)

1. Define and Elaborate agents in AI with examples (any two)?
2. What is expert systems? Elaborate your answer with two examples?
3. What is uncertainty? Briefly describe the fuzzy logic in predicting uncertainty?
4. Present a comprehensive comparative analysis of different supervised learning techniques?
5. Briefly describe the concept of pruning? why and where it is implemented in AI?

Q # 2 Solve it with the help of any unsupervised learning technique. Assume Centre points are (in case of solving thorough K-means clustering): (1,1) and (5,7) (marks 5)

subject	A	B
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

Pass 1:
1 - A
2 - B
3 - A
4 - C
5 - D
6 - B
7 - C
8 - D
9 - B
10 - C
Pass 2:
1 2 3 4 5 10 9 7 6 8
Pass 3:
1 2 3 4 5 6 7 8 9 10
15, 25, 28, 30, 35, 40, 45, 50, 55, 60, 70

Q # 3 Apply chaining techniques on given facts based on following rules

(marks 5)

Rules:

R1: if hot AND smoky Then ADD fire

R3: If fire then Add switch_on_sprinklers

R5: If Sprinklers_on Then DELETE dry

Facts:

F1: alarm_beeps

F2: hot

Goal: Should I switch sprinklers on?

R2: If alarm_beeps Then Add smoky

R4: If dry Then ADD switch_on_humidifier

Q # 4 Implement A* search algorithm on the following graph and find the best route starting from S to Goal G. (marks 5)

